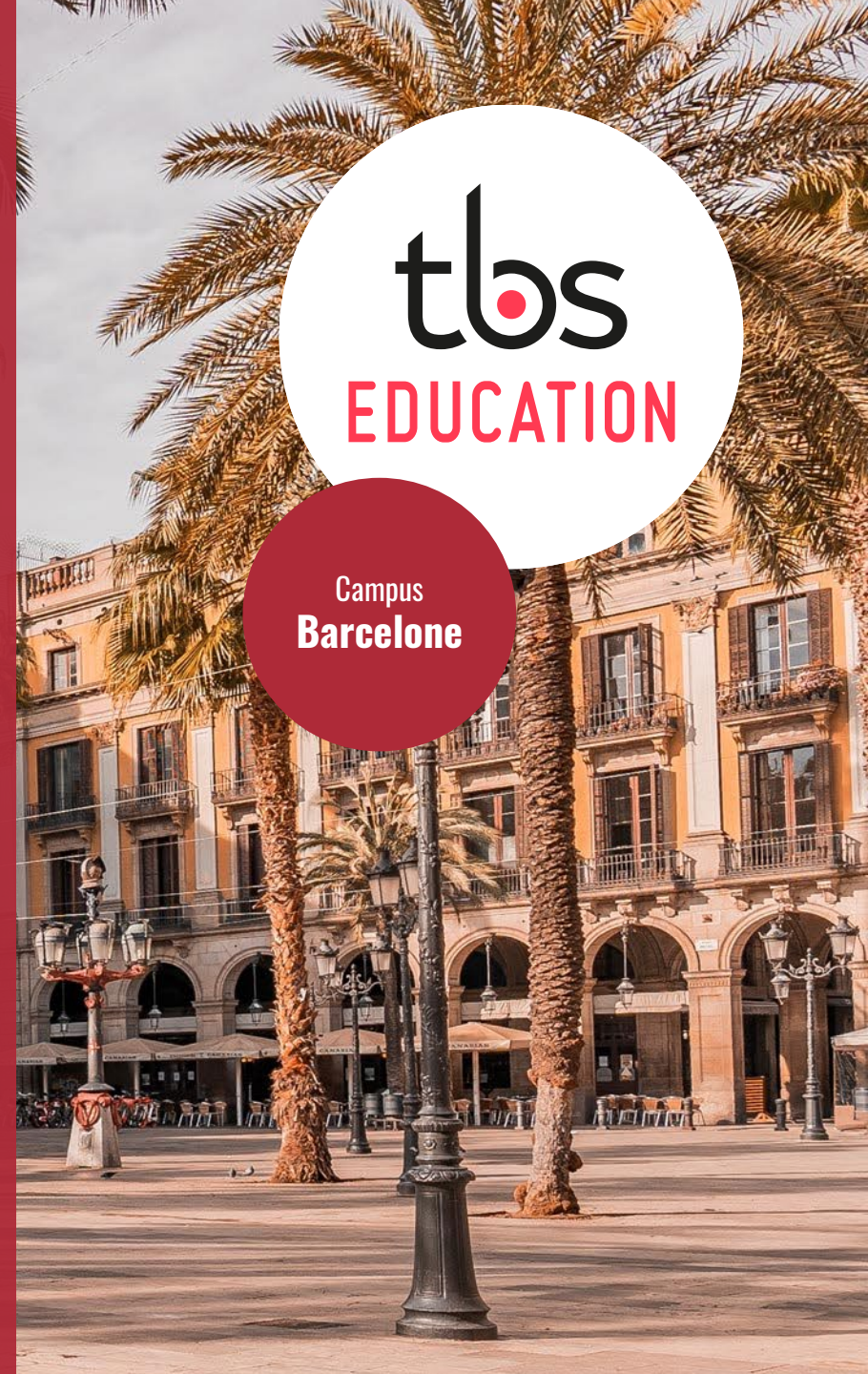


The Business Aspects of Artificial Intelligence 2026

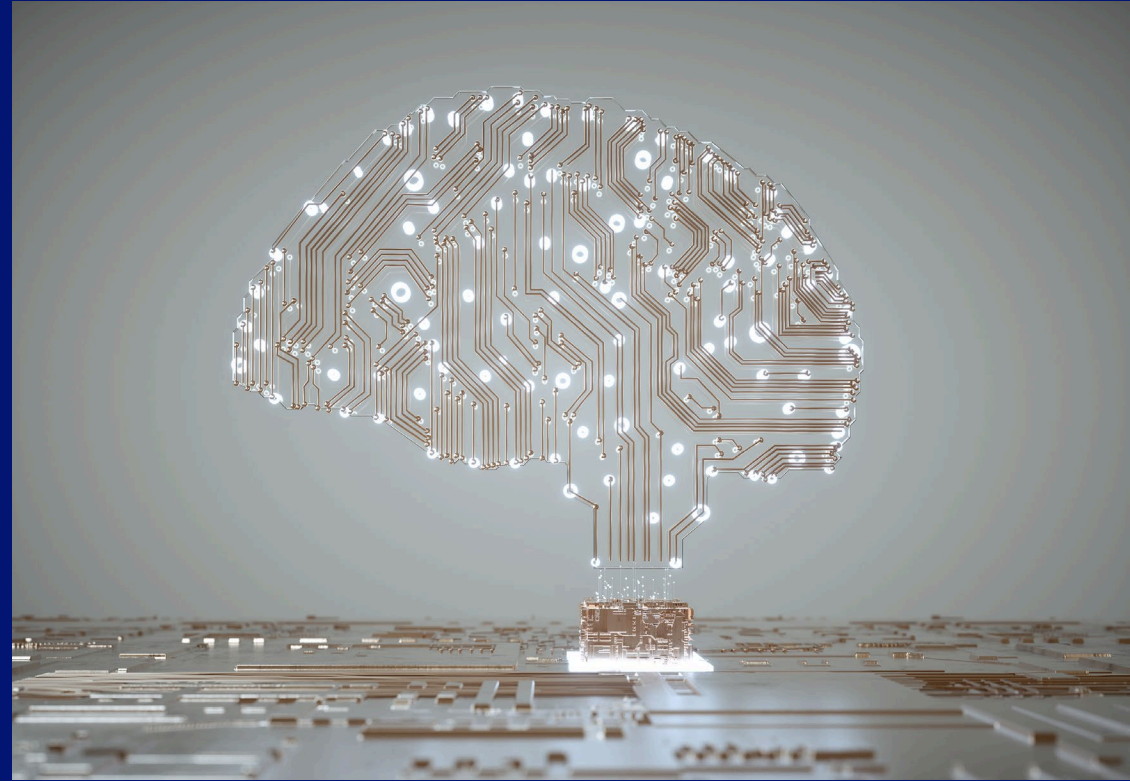
Daniel Martin PhD

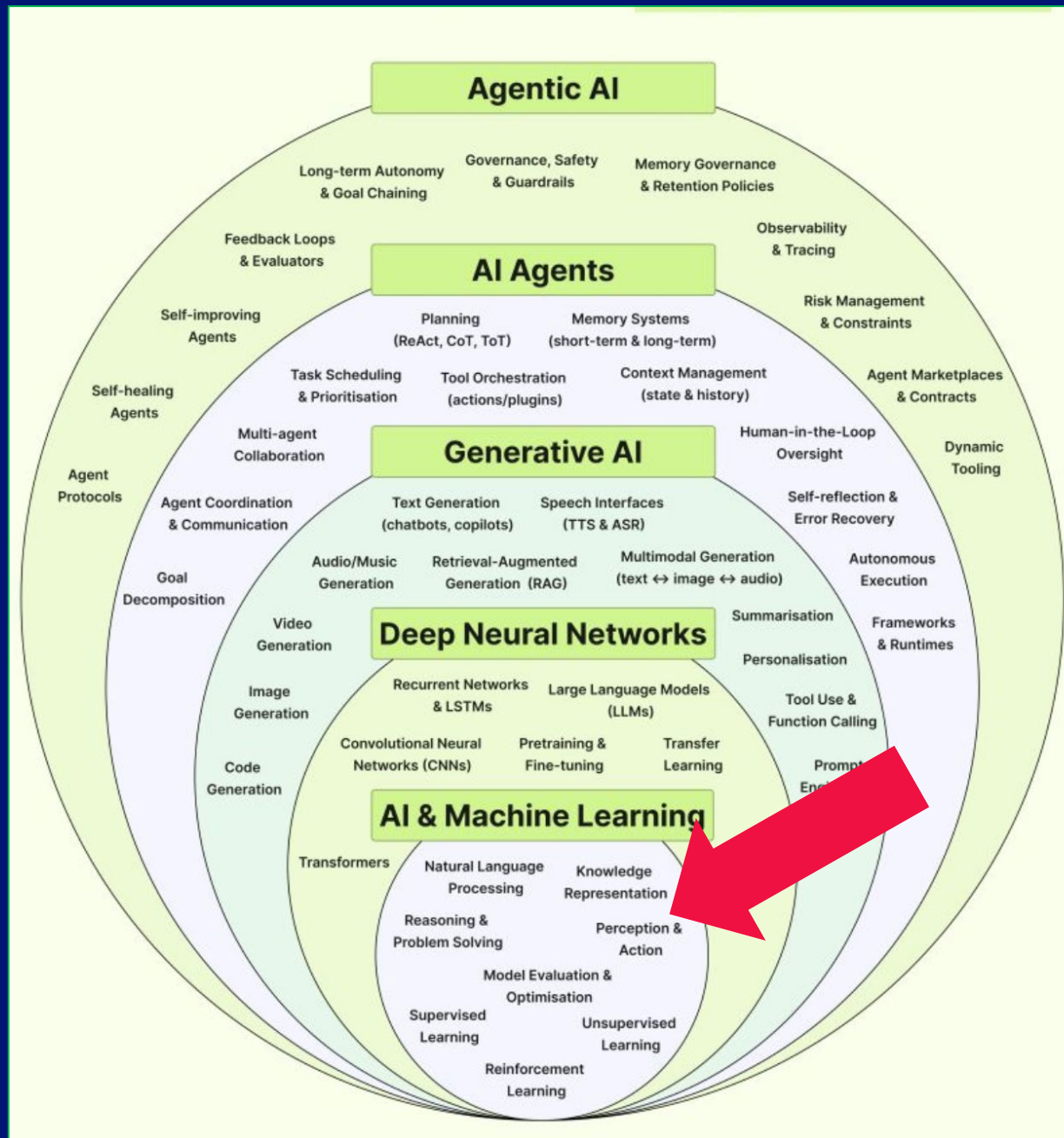


AN INTRODUCTION TO
MACHINE LEARNING



S1: Machine Learning





Source: EndGame, 2026

Context: Things are about to get “technical”. However, we don’t require you to learn all the detail such as matrices or equations but we do expect you to be able to “tell the story” (with credibility), answer basic questions and display “intuition” for the topic and recall high school topics i.e. calculate the gradient of a line on a graph & know the equation...!



Machine Learning (ML): Background

- **Machine Learning (ML)** is a subfield of Artificial Intelligence that focuses on building systems that learn from data to identify patterns and make decisions with minimal human intervention.
- In the early days of AI (the 1950s and 60s), researchers tried to achieve intelligence by hard-coding every possible rule into a computer. However, they soon realized that human-level complexity is too vast for manual programming. This led to the emergence of ML, which flipped the script: instead of humans telling the computer what to do, we gave the computer data and let it "learn" the rules itself.

Key Historical Milestones

- **1952 – The First Learner:** Arthur Samuel created a "learning" checkers-playing program.
- **1959 – Coining the Term:** Samuel officially coined the phrase "Machine Learning".
- **The 1990s – The Data Revolution:** ML shifted from a purely academic pursuit to a data-driven science.
- **The 2010s – The Deep Learning Boom:** The rise of massive datasets (Big Data) and specialized hardware (GPUs) allowed ML to transition into "Deep Learning".

Machine Learning (ML): Core Components

1. **The Data:** Data is the foundation. Without it, the machine has nothing to learn from.

- **Features:** These are the input **independent variables** (e.g., in real estate, features would be square footage, number of bedrooms, and location).
- **Labels:** This is the "answer/**dependent variable**" the machine is trying to predict (e.g., the final sale price of the house).
- **Training vs. Testing Sets:** Data is usually split. You use the **training set** to teach the model and the **testing set** to see if it learned correctly but this is subject to sampling errors.

2. **The Model:** The model is the mathematical representation of the real-world process. The internal settings are called **parameters**.

- **The Algorithm:** This is the specific mathematical approach used (e.g., Linear Regression, Decision Trees, or Neural Networks).

Note: How the algorithms are based on the statistical techniques that have been "around" for a long time! So, they are subject to same

limitations!

- **The Hypothesis:** As the model learns, it forms a hypothesis about how the inputs (features) relate to the outputs (labels).

3. **The Objective Function (The Goal):** We/it need to know if it is doing a good job.

- **Loss Function (Error):** This measures how far the model's prediction was from the actual truth. If the model predicts a house is worth \$200k but it sells for \$250k, the loss is high.

- **Optimization:** The "learning" part of ML is just the machine trying to minimize that loss function by adjusting its internal parameters until the errors are as small as possible.

4. **Quality indicators:** Quantitative indicators of quality of the model. Such as Precision

Machine Learning (ML): Types of ML

How a machine learns/operates depends on the "supervision" it receives:

Supervised Learning: The machine is given "labelled" data (variables). It's like a student learning with an answer key.

- *Examples:* Spam detection, image recognition, price forecasting.

Unsupervised Learning: The machine is given data but no labels. It must find its own patterns or structures.

- *Examples:* Customer segmentation (grouping similar shoppers), anomaly detection (finding fraudulent bank transactions).

Reinforcement Learning: The machine learns through trial and error. It receives "rewards" for good actions and "penalties" for bad ones.

- *Examples:* Self-driving cars, robots learning to walk, or AI playing video games.



Machine Learning (ML): Pipelines

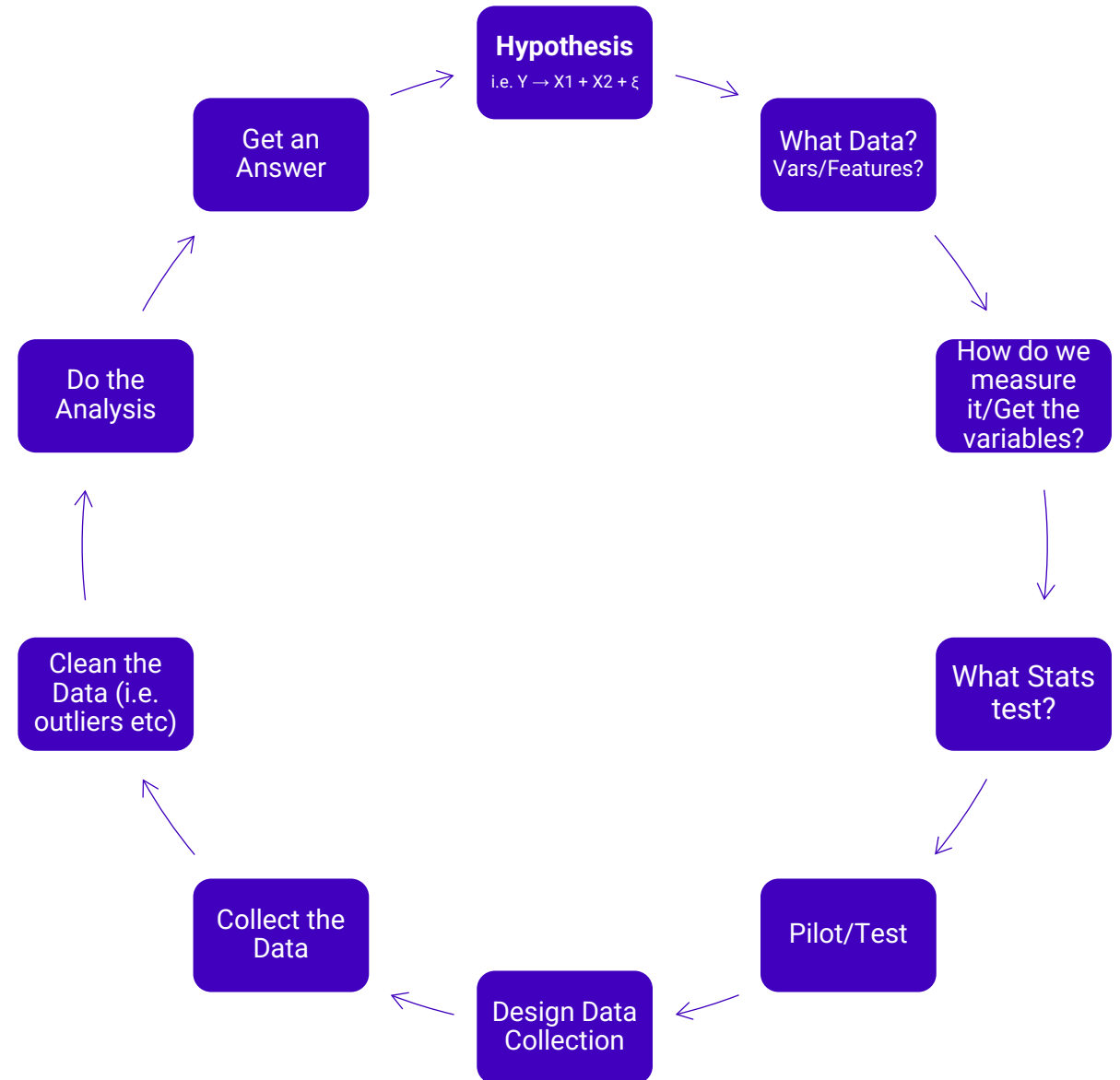
The ML Pipeline (How it works in practice):

- **Data Collection:** Gathering the raw information. (ETL/ELT)
- **Feature Engineering:** Selecting and cleaning the most important pieces of data.
 - Deductive (Supervised learning)
 - Inductive (unsupervised learning)
- **Model Selection:** Choosing the right algorithm for the task (analysis).
- **Training:** Running the data through the algorithm to minimize error.
- **Evaluation:** Testing the model on data it has never seen before to ensure it works



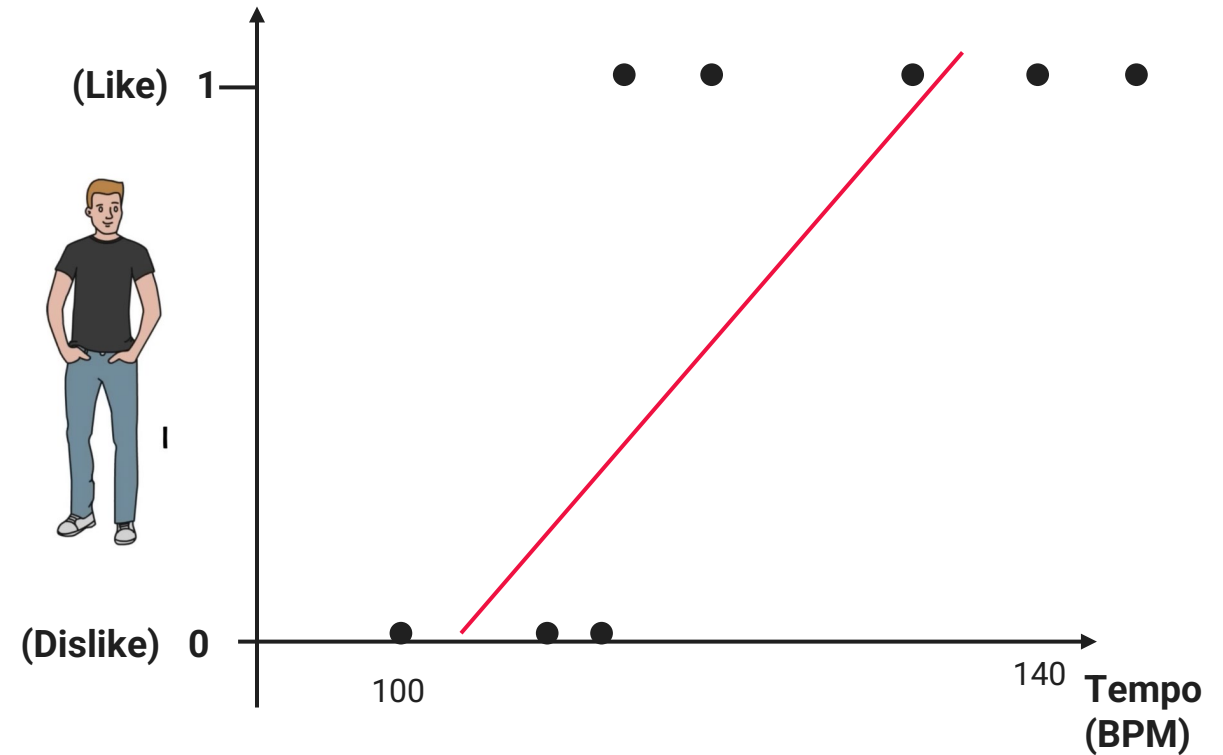
Machine Learning (ML): Research Design is Still Important..!

- Because ML relies on a statistical test – the normal considerations for research design apply!
- If not, you will get an answer but not a credible one!
- If you don't know this then you might mislead yourself!
- But many people don't do these.
- Many core parts of AI are “broken” for these reasons.



Machine Learning (ML): A Basic Algorithm

- John Likes music and we are going to conduct some research with John.
 - We predict that young males in BCN prefer music with higher tempo (BPM) - a priori hypothesis = deduction.
 - We collect some data i.e. play john different songs.
 - What is the mathematical equation of that line (high school stuff)?
 - $y = mx + b$
- **y:** The predicted output (the vertical axis representing Like/Dislike).
 - **x:** The input feature (Tempo in BPM).
 - **m:** The slope i.e. the weight or importance the model assigns to the tempo.
 - **b:** The y-intercept, which is the value of y when the tempo is zero.
 - **i.e. $y = 0.01x + 0$** (Linear Regression - Ordinary Least Squares (OLS) – estimator)
 - **What m represents in ML?** m is called the weight(ing) & it tells the model how much "importance" to give to the Tempo. A small positive value tells the model: "As tempo increases, the probability of a 'Like' increases slightly."



To turn that straight red line into a curve that actually fits the "binary" (0 or 1) nature of your data, we use the **Logistic Regression** equation.

Instead of just calculating $y = mx + b$, we wrap that linear result inside a **Sigmoid Function**. This "squashes" any value into a range between 0 and 1, creating the characteristic S-curve.

The Equation

The full equation for the probability P of a song being "Liked" is:

$$P = \frac{1}{1 + e^{-(mx+b)}}$$

Machine Learning (ML): A Basic Algorithm

- Can we make predictions from the straight line (OLS)?
- Yes but accuracy is an issue in a certain range of X (Tempo) – Purple zone

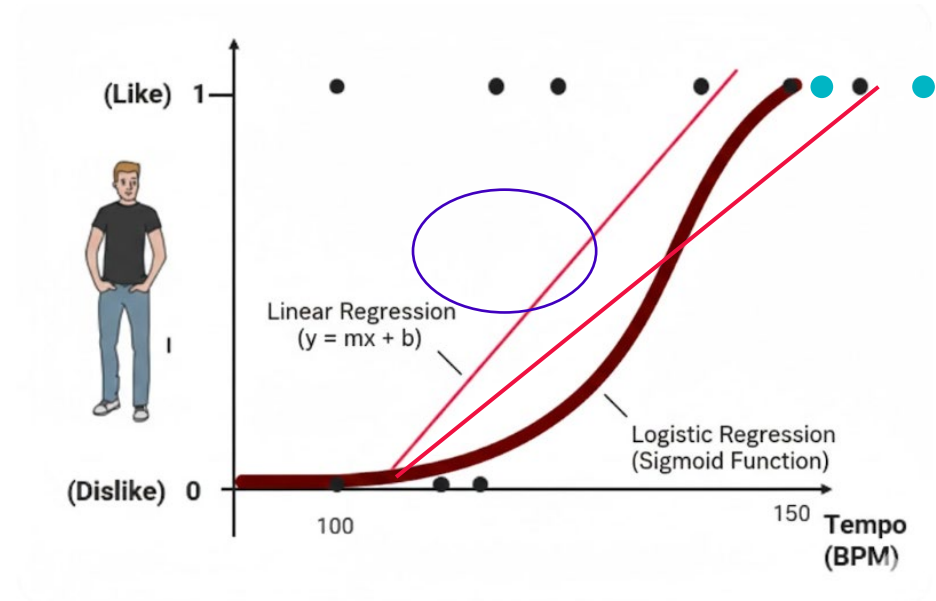
To turn that straight red line into a curve that actually fits the "binary" (0 or 1) nature of your data, we use the **Logistic Regression** equation.

Instead of just calculating $y = mx + b$, we wrap that linear result inside a **Sigmoid Function**. This "squashes" any value into a range between 0 and 1, creating the characteristic S-curve.

The Equation

The full equation for the probability P of a song being "Liked" is:

$$P = \frac{1}{1 + e^{-(mx+b)}}$$



- As we get more data the algorithm (if recalculated) will compute a new m & b and hence a new line and/or curve.
- Has the machine/computer learned?
- We are applying/using a logistic regression (statistical model).
- Can we make predictions? Yes!
 1. Will john like a song with 180 BPM?
 2. Below what BPM is john likely not to like a song?

Machine Learning (ML): A Basic Algorithm

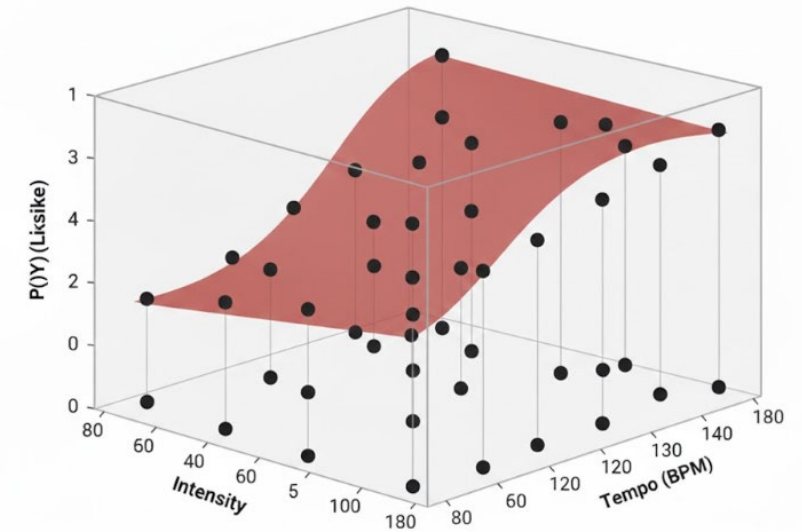
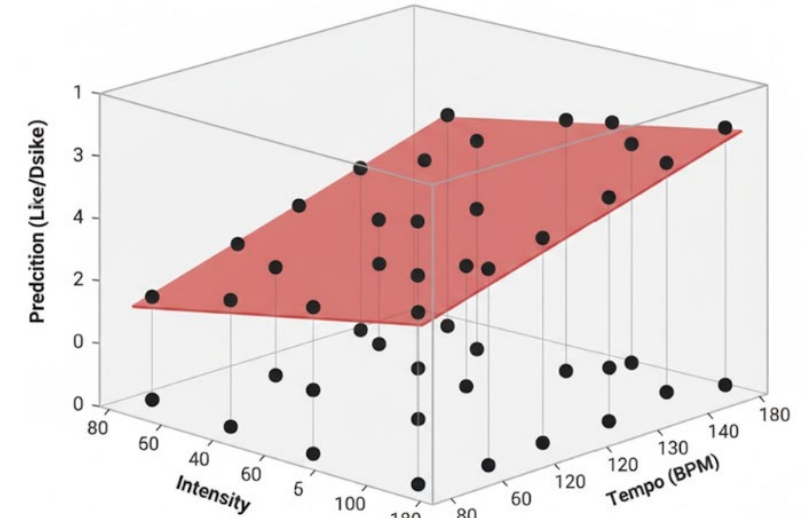
- When we move from one variable (Tempo) to two variables (**Tempo** and **Intensity**), the chart changes from a 2D line to a **3D Plane**.
- "Like/Dislike" outcome is no longer a point on a line, but a height on a flat surface tilted in 3D space.
- The mathematical formula expands to include the second variable/feature:

$$y = m_1x_1 + m_2x_2 + b$$

- y : Like/Dislike score (0 to 1).
- x_1 : Tempo (BPM).
- x_2 : Intensity (how "loud" or "heavy" the song is).
- m_1 : The weight/slope for Tempo.
- m_2 : The weight/slope for Intensity.
- b : The base "Like" level (y-intercept).

Y (Like)	Tempo	Intensity	?
1	160	60	
0	123	90	
1	152	70	

- Can you visualise more dimensions/features?



ML: Video

<https://duckduckgo.com/?q=Machine+learning+basics&t=chromentp&ia=videos&iax=videos&iai=https%3A%2F%2Fwww.youtube.com%2Fwatch%3Fv%3DukzFI9rgwfU>

Machine Learning (ML): Algorithms

1. Regression Algorithms

- Used for predicting a continuous numerical value (e.g., house prices, stock trends, or temperature). **Linear Regression:** The most basic model. It assumes a straight-line relationship between the input variables and the output.
- Polynomial Regression:** Used when the relationship between data points is curved rather than a straight line.
- Ridge / Lasso Regression:** Variants of linear regression that use "regularization" to prevent the model from becoming too complex and over-fitting the training data.
- Decision Tree Regression:** Breaks data down into smaller and smaller subsets (branches) based on specific numerical thresholds to predict a value.

2. Classification Algorithms

- Used for predicting a discrete category or label (e.g., Spam vs. Not Spam, Cat vs. Dog).
- Logistic Regression:** Despite the name, this is for classification. It uses a Sigmoid

function to predict the probability of an item belonging to a specific class (usually 0 or 1).

Support Vector Machines (SVM): Finds the optimal "boundary" (hyperplane) that separates different classes with the widest possible margin.

Naive Bayes: A probabilistic classifier based on Bayes' Theorem. It is exceptionally fast and commonly used for text classification (like spam filters).

K-Nearest Neighbours (KNN): Classifies a data point based on how its "neighbours" are classified. If most of the closest points are "Blue," the new point becomes "Blue".

Random Forest: An "ensemble" method that combines hundreds of individual Decision Trees and takes a majority vote to decide the final classification. This is highly accurate and robust.

3. Clustering Algorithms (Unsupervised)

- Used when your data has no labels. The machine must find hidden patterns and group similar items together.

K-Means Clustering: Groups data into "\$K\$" number of clusters. It works by finding the centre (centroid) of each group and assigning points to the nearest one.

Hierarchical Clustering: Builds a tree-like structure of clusters. You can "cut" the tree at different levels to get more or fewer groups.

DBSCAN: A density-based algorithm. Unlike K-Means, it can find clusters of any shape and is excellent at identifying "noise" or outliers in the data.

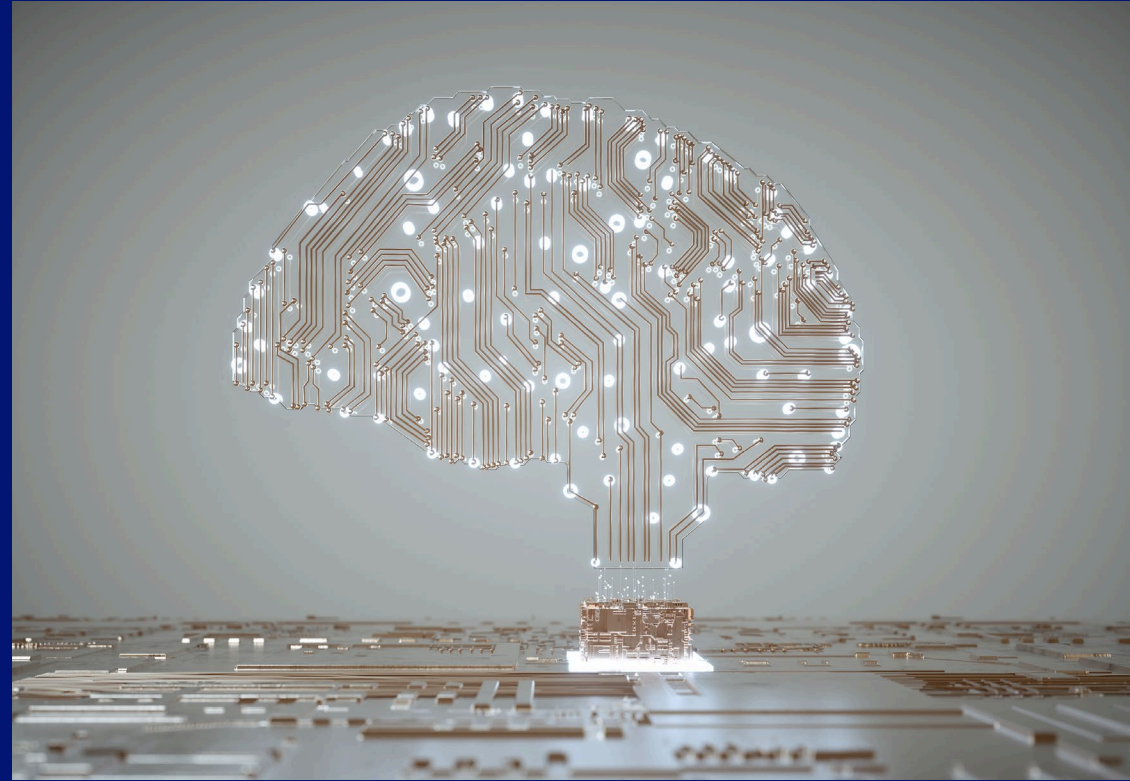
4. Dimensionality Reduction

These aren't for prediction, but for simplifying data before running other algorithms.

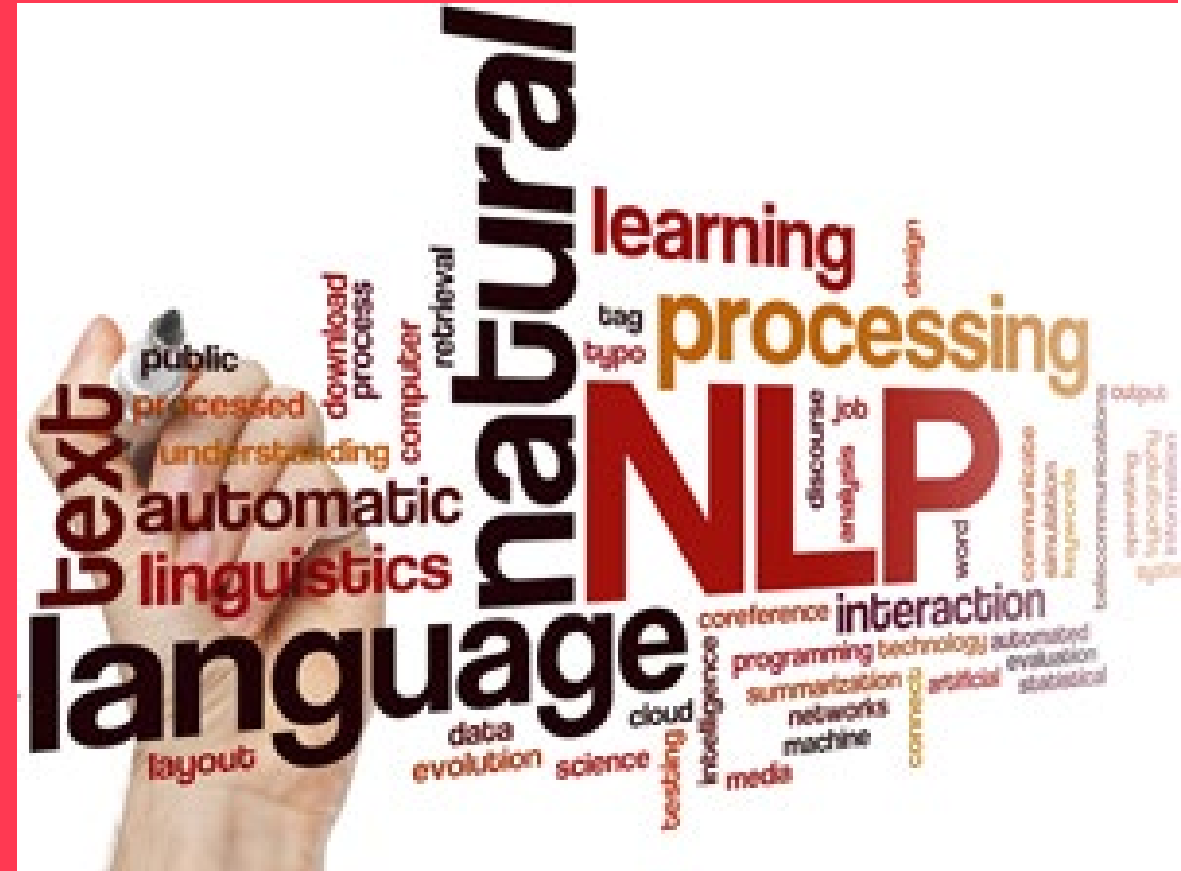
Principal Component Analysis (PCA): Reduces the number of variables in a dataset while keeping as much information as possible. It is like "compressing" a 3D image into 2D without losing the main shapes.

Each has "well known" limitations...! Well known to statisticians in any case!

S1: Natural Language Processing



What is Natural Language Processing (NLP)?



Natural Language Processing(NLP): The Basics

1. What is NLP?

Natural Language Processing is the intersection of **Linguistics**, **Computer Science**, and **Artificial Intelligence**. Its goal is to bridge the gap between human communication (unstructured, ambiguous) and computer data (structured, numeric).

- **The Core Challenge:** Human language is messy. It relies on context, sarcasm, idioms, and evolving grammar, which are difficult for traditional "if-then" logic to handle.

2. The Evolution of NLP

- **Rule-Based (1950s–1990s):** Hand-written grammar rules and dictionary lookups.
- **Statistical (1990s–2010s):** Using probability to guess the next word based on previous data.
- **Deep Learning (2014–Present):** Using **Neural Networks** and **Transformers** to understand context and relationships between words.

3. Core Components of the NLP Pipeline

Before a machine can "read," the text must be pre-processed into a format it understands.

- **Tokenization:** Breaking sentences into individual words or "tokens."
- **Stop Word Removal:** Filtering out common words that add little meaning (e.g., "and," "the," "is").

- **Stemming/Lemmatization:** Reducing words to their root form (e.g., "running" becomes "run").
- **Vectorization (Embeddings):** Converting words into numbers. Modern NLP uses **Word Embeddings**, where words with similar meanings are mathematically "close" to each other in a multi-dimensional space.

4. Key NLP Tasks:

NLP is generally divided into two main categories:

NLU (Natural Language Understanding): Focuses on "reading" and comprehension.

- **Sentiment Analysis:** Is this review positive or negative?
- **Named Entity Recognition (NER):** Identifying people, places, or dates in a text.
- **Topic Modelling:** Automatically categorizing articles into themes.

NLG (Natural Language Generation): Focuses on "writing" and creating human-like text.

- **Machine Translation:** Converting English to Spanish (e.g., Google Translate).
- **Summarization:** Condensing a long document into a few bullet points.
- **Dialogue Systems:** Conversational agents and LLMs (e.g., ChatGPT).

Large Language Models (LLM): The Basics

1. What is an LLM?

- A Large Language Model is a type of AI trained on massive amounts of text to understand, summarize, generate, and predict human language.
- **Large:** Refers to both the **dataset** (trillions of words from the internet, books, and code) and the **parameters** (billions of internal "switches" that define the model's knowledge).
- **Language:** The core focus is on natural communication—not just grammar, but context, tone, and reasoning.
- **Model:** It is a complex mathematical representation of how humans use language to convey meaning.

2. How LLMs Work: The "Next Token" Prediction

- At its most basic level, an LLM is a high-speed prediction engine. It doesn't "know" facts; it calculates the **probability** of the next word.

Input: "The capital of France is..." **Model Calculation:**

- Paris (98% probability)
- Lyon (1% probability)
- London (0.5% probability) **Output:** "Paris"

3. The Engine: The Transformer

- Modern LLMs (like GPT-4 or Llama 3) use the **Transformer** architecture. Its secret weapon is **Self-Attention**:

- Older AI read text one word at a time (left-to-right).
- Transformers look at the **entire sentence at once**, allowing them to understand that in the phrase "*The bank of the river*," the word "river" completely changes the meaning of "bank."
- They look at the sentence in many ways (heads/multi-heads).

4. Risks & Limitations (The "Reality Check")

It is vital to present these alongside the benefits:

Hallucinations: The AI can state false information with total confidence because it is predicting words, not verifying facts.

Bias: If the training data contains human prejudices, the model will likely repeat them.

Context Window: Models have a "memory limit" for how much text they can consider at one time.

Data Privacy: Information shared with public LLMs may be used for future training.

Large Language Models (LLM): Token

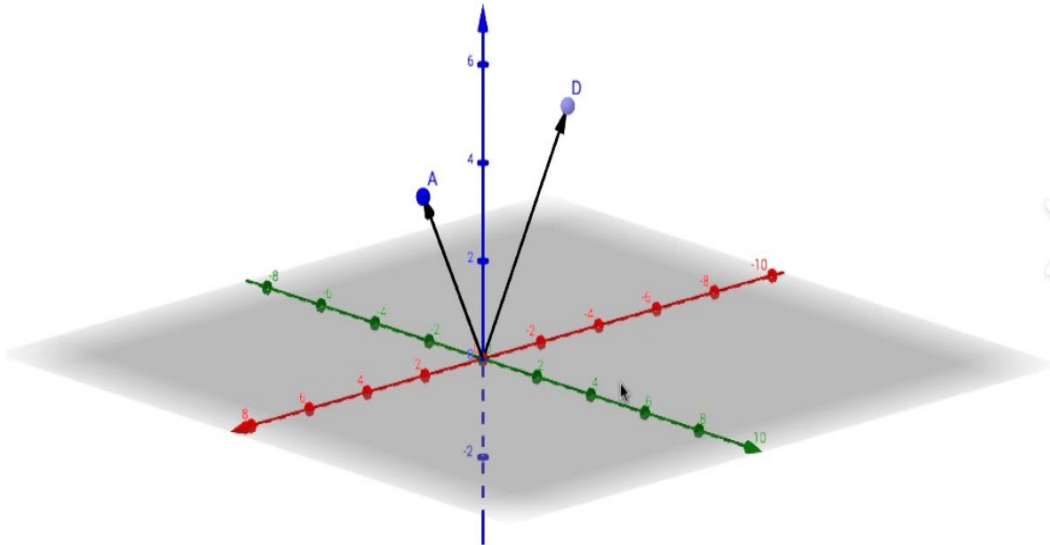
➤ Example of an Embedding Matrix Lookup Table (Vectors)

Word	Token	Animal	Furry	Large
Elephant	4457	1.0	0.1	0.8
Tiger	1023	1.0	1.0	0.65
Frog	867	1.0	0.0	-0.2

- 4457 = [1.0, 0.1, 0.8]
- 1023 = [1.0, 1.0, 0.65]
- 867 = [1.0, 0.0, .0,2]

LLM Intro Video:

<https://duckduckgo.com/?q=H0w+do+LLM%C2%B4s+work%3F&t=chromentp&ia=videos&iax=videos&iai=https%3A%2F%2Fwww.youtube.com%2Fwatch%3Fv%3DPaNPWuAS0vM>



Positional Encoding

I	went	outside	to	plant	a
1	2	3	4	5	6

Noun (Subject) Encoding

I	went	outside	to	plant	a
6	3	4	2	5	1

Topic (Gardening) Encoding

I	went	outside	to	plant	a
3	4	5	2	6	1

LLMs: The Transformer

Modern Architecture: The Transformer

The current "state of the art" in NLP is the **Transformer model**. Unlike older models that read text left-to-right, Transformers use a mechanism called **Self-Attention**.

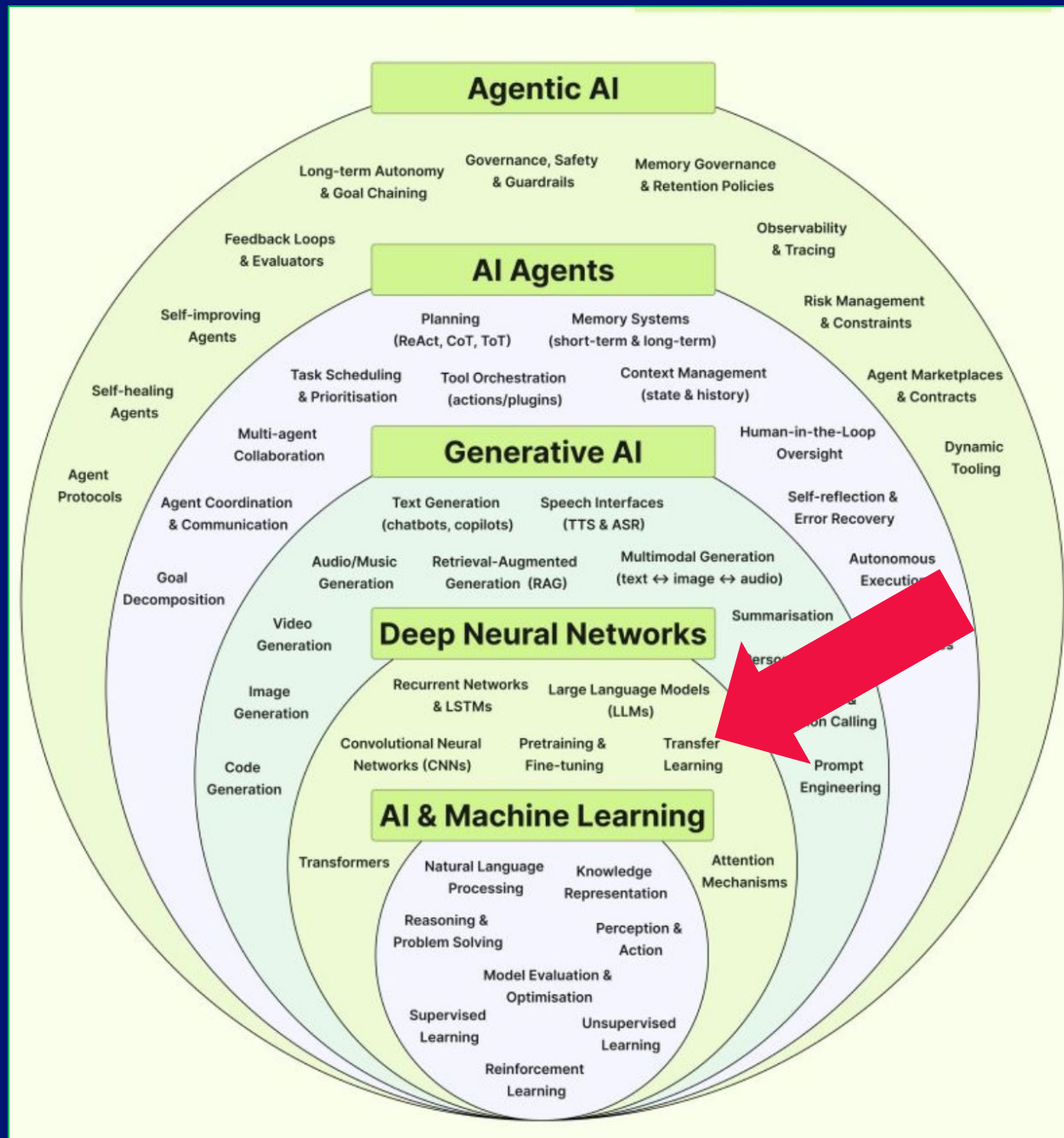
Self-Attention: Allows the model to look at every word in a sentence simultaneously to determine which words are most relevant to each other (e.g., in "The bank of the river," the word "river" tells the model that "bank" isn't a financial institution).

Real-World Applications

- **Search Engines:** Semantic search (finding intent, not just keywords).
- **Virtual Assistants:** Siri, Alexa, and Google Assistant.
- **Content Moderation:** Automatically flagging hate speech or spam.
- **Healthcare:** Extracting patient data from unstructured doctor notes.

VIDEO (LLMs):

<https://duckduckgo.com/?q=what+does+transformer+do+in+llm&t=chromentp&ia=videos&iax=videos&iai=https%3A%2F%2Fwww.youtube.com%2Fwatch%3Fv%3DeMlx5fFN0Yc>



Source: EndGame, 2026

Deep Neural Networks (DNN):

Deep Neural Networks (DNNs) are the "brain" behind modern AI. They are designed to mimic the way human neurons pass signals to one another to recognize patterns.

1. What makes it "Deep"?

A basic Neural Network has an input and an output. A **Deep** Neural Network has multiple **Hidden Layers** stacked between them. These layers allow the network to learn increasingly complex features:

1. **Simple Layer:** Recognizes edges or lines.
2. **Deep Layer:** Recognizes shapes or textures.
3. **Final Layer:** Recognizes complex objects (like a human face or a car).

2. How it Works: The Mathematical Path

Weights & Biases: Each connection between "neurons" has a weight. This determines how much importance to give to a specific input.

Activation Function: Each neuron uses a mathematical function (like

ReLU or Sigmoid) to decide whether the signal is "strong enough" to be passed to the next layer.

Backpropagation: This is how the network learns. It compares its guess to the correct answer, calculates the error, and goes backward through the layers to adjust the weights.

3. Key Architectures to Know

Different "shapes" of networks are used for different tasks:

CNNs (Convolutional Neural Networks): Best for **Vision**. They "scan" images like a flashlight to find patterns.

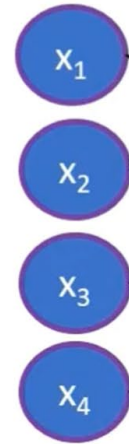
RNNs (Recurrent Neural Networks): Best for **Sequences**. They have "memory," making them ideal for time-series data or basic speech.

Transformers: The modern standard. They use "Attention" to process massive amounts of data in parallel (the foundation of LLMs).

<https://duckduckgo.com/?q=depp+enural+network&t=chomentp&ia=videos&iax=videos&iai=https%3A%2F%2Fwww.youtube.com%2Fwatch%3Fv%3DZBnHoVMeeJE>

**Features/Independent
Vars:**
i.e. GDP,
Av. National Inc etc.

$$x_{\text{kiribati}} = \begin{bmatrix} 188 \\ 511 \\ 4410 \\ 2 \end{bmatrix}$$



INPUT LAYER

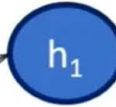
"like" a Linear
Regressions/Sigmoid
function to give "weights"



a_1

a_2

a_3



**Factors: i.e. Socio-
economic.**

HIDDEN LAYER



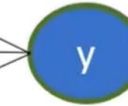
b_1

b_2

b_3

MATRICES...!

**Labels: i.e.
Output/Dependent Vars**

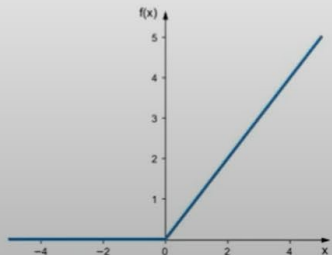


Life
Expectancy

"like" Nodes/Neurons but just
variables with values i.e.
datapoints

OUTPUT LAYER

$$\mathcal{F}(x) = \text{ReLU}(x)$$



$$y = \mathcal{F}_2(\mathbf{B}\mathcal{F}_1(\mathbf{A}\mathbf{x}))$$

$$\mathbf{h} = \mathcal{F}_1(\mathbf{A}\mathbf{x})$$

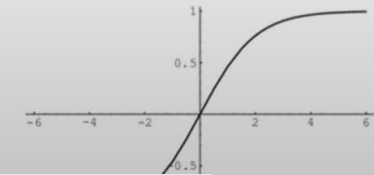
$$y = \mathcal{F}_2(\mathbf{B}\mathbf{h})$$

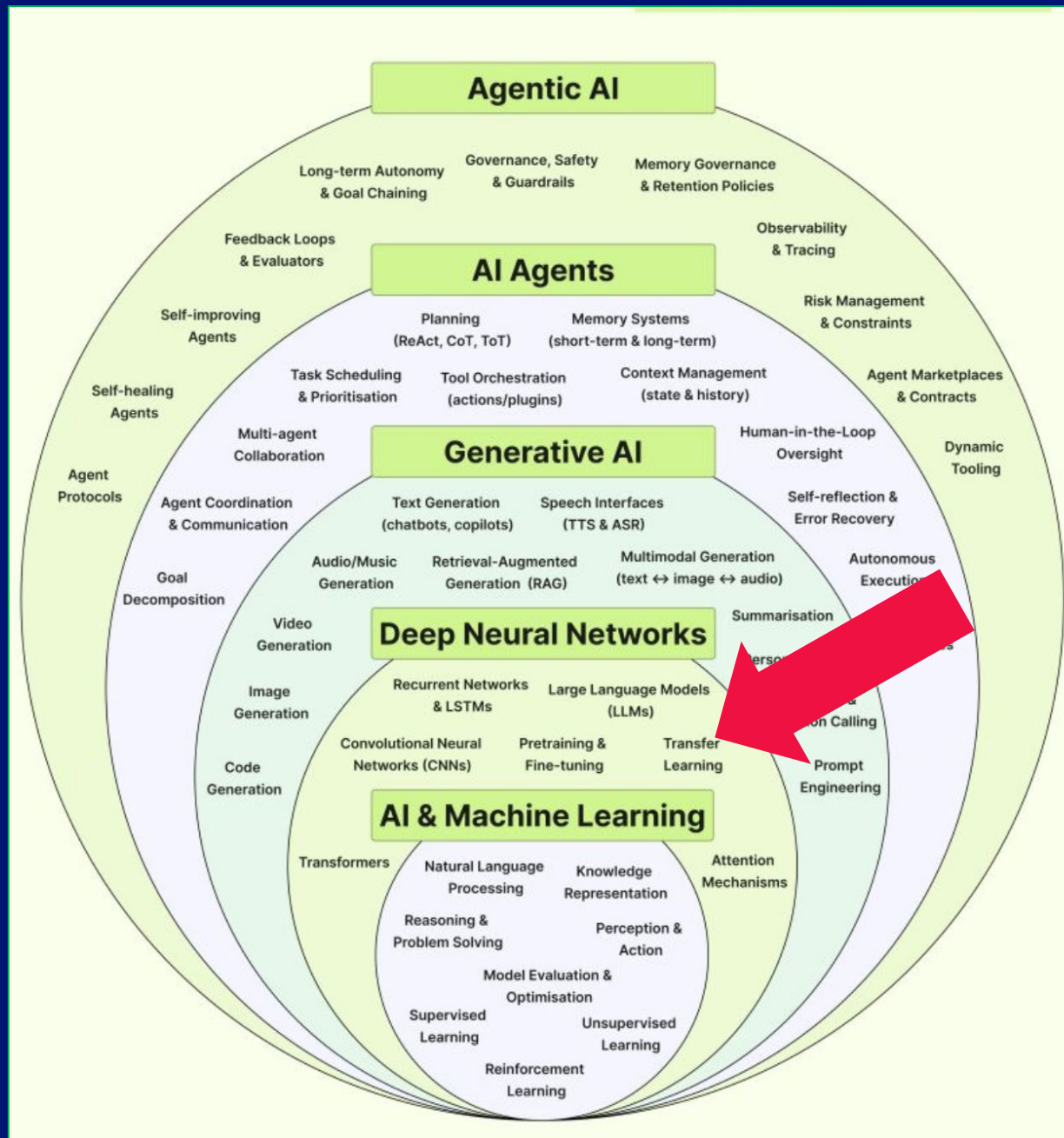
$$h_1 = A_{11}x_1 + A_{12}x_2 + A_{13}x_3 + A_{14}x_4$$

$$h_2 = A_{21}x_1 + A_{22}x_2 + A_{23}x_3 + A_{24}x_4$$

$$h_3 = A_{31}x_1 + A_{32}x_2 + A_{33}x_3 + A_{34}x_4$$

$$\mathcal{F}(x) = \tanh(x)$$





Source: EndGame, 2026



Thank you!

L'école de management triplement accréditée



[tbs-education.fr](https://www.tbs-education.fr)